

CS201 Midterm 2

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Abstract— Write only ONE function per question. Do not check the input, assume that the input is correct. Unless noted,

- (i) The input data structure is not empty
- (ii) Time complexity of the algorithm does not matter
- (iii) You can not use extra data structures including arrays
- (iv) You can use any function given in the data structure
- (v) You can not modify the data structure contents

I. QUESTION (TREE)

Write the recursive method

```
int[] lessThanX(int x)
```

in `TreeNode` class which collects all values in all nodes in the tree. The size of the array should be as large as it should be. (Hint: First calculate how many items the array should contain, then write the same code to fill in the array. The left and right child function calls should return arrays, and the method should concatenate them)

II. QUESTION (TREE)

Write a non-recursive method

```
void bottomTwo()
```

which prints the two smallest numbers in the binary search tree. You are not allowed to use any external data structures. (Hint: There are 3 cases: (i) The smallest number is on the left side and it has right descendants, (ii) The smallest number is on the left side and it has no right descendants, (iii) The smallest number is the root.)

III. QUESTION (QUEUE)

Write the method

```
int maximum()
```

returns the maximum number in a queue. Write the method for the linked list implementation.

IV. QUESTION (QUEUE)

Write the method

```
void thisMustChange(int[] indexOfNonbribers)
```

which simulates how the queue goes in an hypothetical country, which we all aware of. *indexOfNonbribers* is an increasing order sorted array showing the indexes of the people who have not bribed the officer in the queue. The method will send these people to the end of the queue in their respective order. The first element in the queue has the index 0. Write the method for the linked list implementation.

V. QUESTION (HASHING)

Write the static method

```
int [] symmetricDiff(int[] list1 , int[] list2 )
```

to find the symmetric difference of the elements in two arrays $(A - B) \cup (B - A)$ and return a new array. The symmetric difference array should contain only that many items not more not less. Your method should run in $\mathcal{O}(N)$ time, where N is the number of elements in the arrays. Do not use any external data structures or arrays except the resulting array and two external hash tables. Write the method for the array implementation.

VI. QUESTION (HASHING)

Write the method

```
int minTableSize(int[] numbers)
```

which identifies the minimum hash table size, where there will be no collisions at all. Use an external linked list based hash table for checking collisions.